Homework 2

NUMA01: Computational Programming with Python

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This assignment has 11 tasks.
All the functions must be *properly documented* and also tested when possible. Instructions on how to hand in and how to present are given on the corresponding homework page of the course platform.

Interval analysis

Theory

Interval analysis is a branch of mathematics that deals with the theory of intervals. Instead of e.g. real numbers, the basic objects in interval analysis are closed intervals $[a, b]$.

The motivation for this is that in the real world, measurements are usually not exact, but come with some error margin. So the mass of an object, for example, might be $m = 3 \pm 10^{-3}$ kg, if the scale has a sensitivity of 1 g. This is naturally expressed as $m \in [3 - 10^{-3}, 3 + 10^{-3}]$ kg. If we also measure the acceleration of this object, $a \in [2 - 10^{-2}, 2 + 10^{-2}]$ m/s^2, then by Newton's second law $F = ma$ it follows that the force F exerted on the mass lies in the interval

$$F \in [(3 - 10^{-3})(2 - 10^{-2}), (3 + 10^{-3})(2 + 10^{-2})].$$

Note how the lower limit on the measurement error is less than the upper limit, so we cannot write $F = 6 \pm \text{err}$ without losing some information.

We will implement a class for representing intervals and performing operations on them. We consider first the basic arithmetic operations. If $\odot$ is a binary operator (e.g. $+$) then

$$[a, b] \odot [c, d] = \{z \in \mathbb{R} |\ z = x \odot y, \text{where } x \in [a, b] \text{ and } y \in [c, d]\}.$$

This can be written out explicitly as

$$[a, b] \odot [c, d] = [\min(a \odot c, a \odot d, b \odot c, b \odot d), \max(a \odot c, a \odot d, b \odot c, b \odot d)],$$

and for the four basic arithmetic operators we get the following rules:

$$\begin{aligned}
[a, b] + [c, d] &= [a + c, b + d], \\
[a, b] - [c, d] &= [a - d, b - c], \\
[a, b] \cdot [c, d] &= [\min(ac, ad, bc, bd), \max(ac, ad, bc, bd)], \\
[a, b] \,/\, [c, d] &= [\min(a/c, a/d, b/c, b/d), \max(a/c, a/d, b/c, b/d)], \quad 0 \notin [c, d].
\end{aligned}$$

One can also define functions of intervals, though one has to be careful. We just take one example, the power function $x \mapsto x^n$, $n \geq 1$. If n is odd, this is a monotonically increasing function, so we have

$$[a, b]^n = [a^n, b^n]. \tag{1}$$

For even $n > 0$ we have to distinguish between three cases:

$$[a, b]^n = \begin{cases} [a^n, b^n] & a \geq 0, \\ [b^n, a^n] & b < 0, \\ [0, \max(a^n, b^n)], & \text{otherwise.} \end{cases} \tag{2}$$

Be aware that powers of intervals are *not* repeated products,
i.e. $[a, b]^2 \neq [a, b] * [a, b]$.

Task 1

Construct a class Interval which is initialized with two real numbers representing the left and right endpoints respectively.

Task 2

Provide methods for the four basic arithmetic operations.

Task 3

Provide a method on your class so that the code

```
i = Interval(1,2)
print( i )
```

prints

```
[1, 2]
```

Task 4

Make sure that the following code works as expected and prints the values given in the comments

```
I1 = Interval(1, 4)      # [1, 4]
I2 = Interval(-2, -1)    # [-2, -1]
I1 + I2                  # [-1, 3]
I1 - I2                  # [2, 6]
I1 * I2                  # [-8, -1]
I1 / I2                  # [-4.,-0.5]
```

Please note: Use `__truediv__` for the division.

Task 5

Implement the `__contains__` method for checking if a real value is within the given interval.

Task 6

Extend your division function so that it raises appropriate exceptions if the dividing interval contains zero, or if the resulting interval would be infinitely large.

Task 7

A real number r is naturally identified with a degenerate interval $[r, r]$. Extend the class so that it can be initialized with only one real value, i.e.

```
Interval(1)   # [1, 1]
```

Task 8

Modify your code so that the following works

```
Interval(2,3) + 1     # [3, 4]
1 + Interval(2,3)     # [3, 4]
1.0 + Interval(2,3)   # [3.0, 4.0]
Interval(2,3) + 1.0   # [3.0, 4.0]
1 - Interval(2,3)     # [-2, -1]
Interval(2,3) -1      # [1, 2]
1.0 - Interval(2,3)   # [-2.0, -1.0]
Interval(2,3) - 1.0   # [1.0, 2.0]
Interval(2,3) * 1     # [2, 3]
1 * Interval(2,3)     # [2, 3]
1.0 * Interval(2,3)   # [2.0, 3.0]
Interval(2,3) * 1.0   # [2.0, 3.0]
-Interval(4,5)        # see the special method __neg__
```

Task 9

Implement the power function $x \mapsto x^n$ (see equations (1) and (2)) as the `__pow__` function, so that one can write

```
x = Interval(-2,2)   # [-2, 2]
x**2                 # [ 0, 4]
x**3                 # [-8, 8]
```

Task 10

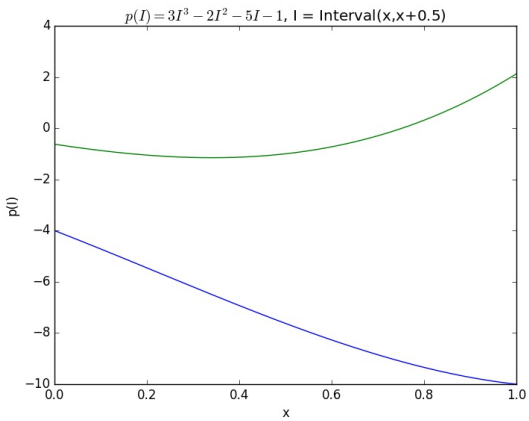
Define a list of 1000 intervals by creating a list of lower boundary values with `xl=linspace(0.,1,1000)` and upper boundaries `xu=linspace(0.,1,1000)+0.5`.

Evaluate the polynomial

$$p(I) = 3I^3 - 2I^2 - 5I - 1$$

on each interval I of your list of intervals and create in such a way another list of intervals. Extract from this lists a list of lower boundaries `yl` and upper boundaries `yu` and plot both versus `xl`.

The result should look like this:



Extra Task 11 (in case you want an extra challenge. Not mandatory!)

Based on your results from Task 3, you may be tempted to extend the interval class for multiple interval values all at once. In the end, `print(interval(1,1))` looks very similar to `print(array([1,1])`. Suppose you have obtained data for some measurement of interval points $[a_i, b_i]$ for $i = 1, 2, ..., N$ stored in arrays `A = array([a_1,...,a_N])` and `B = array([b_1,...,b_N])`.

1. Verify that the arithmetic operations are consistent. Consider multiplying the vectorized interval with some array `C` satisfying `C.shape= (N,N)`. From your knowledge in linear algebra, what result do you expect to obtain?
2. Perform Task 10 by writing interval(xl, xu). Do you get the same plot?
3. Consider Task 10 again. This time, take `xl = linspace(0,1,5 000 000)` and `xu = linspace(0,1,5 000 000) + 0.5`. What is the time difference between running `P(interval(xl,xu))` compared to your original interval class (which only stores a single interval per instance)? Which one is faster? Why?

Good luck!